

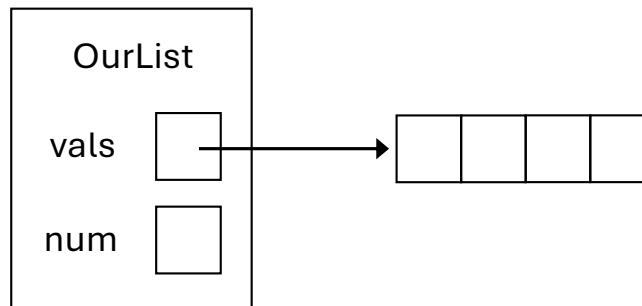
More list implementation

4/7/25

Administrivia

- HW 2 (testing Car) due tomorrow night

Recall: Basic setup for OurArrayList



Which of the following lines has an error?

```
public T remove() {  
    //return arbitrary element (null if List is empty)  
    if(num == 0) //A  
        return null; //B  
    num--; //C  
    T retVal = vals[num]; //D  
    return retVal;  
} //E: Not exactly one of the above
```

Which of the following lines has an error?

```
public T remove() {  
    //return arbitrary element (null if List is empty)  
    if(num == 0) //A  
        return null; //B  
    num--; //C  
    T retVal = vals[num]; //D  
    return retVal;  
} //E: Not exactly one of the above (none)
```

Which of the following lines has a bug?

(E: not exactly one of the above)

```
public boolean remove(T val) {  
    //removes val if it occurs (returns whether it did)  
    for(int i=0; i < num; i++)                //A  
        if(vals[i] == val) {                  //B  
            vals[i] = vals[num-1];            //C  
            num--;                             //D  
            return true;  
        }  
  
    return false;  
}
```

Which of the following lines has a bug?

(E: not exactly one of the above) B and C

```
public boolean remove(T val) {  
    //removes val if it occurs (returns whether it did)  
    for(int i=0; i < num; i++)                //A  
        if(vals[i] == val) {                  //B  
            vals[i] = vals[num-1];            //C  
            num--;                             //D  
            return true;  
        }  
  
    return false;  
}
```

How many of the following correctly implement clear (empties list)?

- | | |
|---|--|
| I. <code>vals.clear();</code> | IV. <code>for(i=0; i<vals.length; i++)</code> |
| II. <code>vals = (T[])new Object[3];</code> | <code>vals[i] = null;</code> |
| III. <code>num = 0;</code> | |

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

How many of the following correctly implement clear (empties list)?

- I. `vals.clear();`
- II. `vals = (T[])new Object[3];`
- III. `num = 0;`
- IV. `for(i=0; i<vals.length; i++)
 vals[i] = null;`

- A. 0
- B. 1 (just III)
- C. 2
- D. 3
- E. 4

Which of the following lines of a List remove method is incorrect?

```
T remove(int i) { //remove and return value at position i
    if((i < 0) || (i > num)) //A
        throw new IndexOutOfBoundsException(); //B
    T retVal = vals[i]; //C
    num--;
    vals[i] = vals[num]; //D
    return retVal;
} //E: Not exactly one of the above
```

Which of the following lines of a List remove method is incorrect?

```
T remove(int i) { //remove and return value at position i
    if((i < 0) || (i > num))                                //A
        throw new IndexOutOfBoundsException();             //B
    T retVal = vals[i];                                     //C
    num--;
    vals[i] = vals[num];                                    //D
    return retVal;
} //E: Not exactly one of the above (A & D)
```